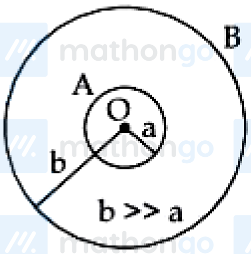
Q1 - 2024 (04 Apr Shift 2)

A rod of length 60 cm rotates with a uniform angular velocity $20 \mathrm{rads}^{-1}$ about its perpendicular bisector, in a uniform magnetic filed $0.5T$. The direction of magnetic field is parallel to the axis of rotation. The potential difference between the two ends of the rod is _____V.

Q2 - 2024 (05 Apr Shift 1)

Two conducting circular loops A and B are placed in the same plane with their centers coinciding as shown in figure. The mutual inductance between them is :



(1) $\frac{\mu_0 \pi b^2}{2a}$
(2) $\frac{\mu_o}{2\pi} \cdot \frac{b^2}{a}$
(3) $\frac{\mu_0}{2\pi} \cdot \frac{a^2}{b}$
(4) $\frac{\mu_0 \pi a^2}{2b}$

Q3 - 2024 (05 Apr Shift 2)

The current in an inductor is given by $\mathrm{I} = (3\mathrm{t} + 8)$ where t is in second. The magnitude of induced emf produced in the inductor is 12mV. The self-inductance of the inductor _____ mH.

Q4 - 2024 (06 Apr Shift 2)

In a coil, the current changes from -2 A to $+2$ A in 0.2 s and induces an emf of 0.1 V. The self inductance of the coil is :

(1) 4mH

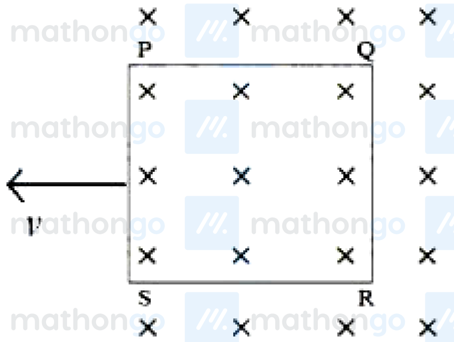
(2) 1mH

(3) 5mH

(4) 2.5mH

Q5 - 2024 (08 Apr Shift 1)

A square loop PQRS having 10 turns, area 3.6×10^{-3} m^2 and resistance 100Ω is slowly and uniformly being pulled out of a uniform magnetic field of magnitude $B = 0.5$ T as shown. Work done in pulling the loop out of the field in 1.0 s is ______ $\times 10^{-6}$ J.

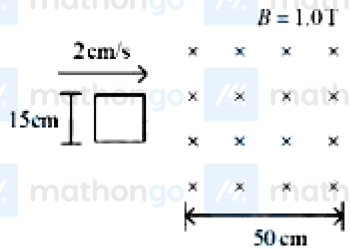


Q6 - 2024 (09 Apr Shift 1)

When a coil is connected across a 20 V dc supply, it draws a current of 5 A. When it is connected across 20 V, 50 Hz ac supply, it draws a current of 4 A. The self inductance of the coil is ______ mH. (Take $\pi = 3$)

Q7 - 2024 (09 Apr Shift 2)

A square loop of side 15 cm being moved towards right at a constant speed of 2 cm/s as shown in figure. The front edge enters the 50 cm wide magnetic field at $t = 0$. The value of induced emf in the loop at $t = 10$ s will be :



(1) 0.3mV

(2) zero

(3) $4.5 mV$

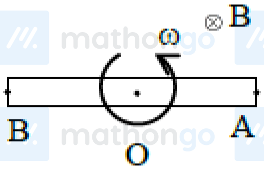
(4) $3 mV$

Answer Key

Q1 (0) **Q2** (4) **Q3** (4) **Q4** (3)

Q5 (3) **Q6** (10) **Q7** (2)

Q1



$$\because V_0 - V_A = \frac{B\omega\ell^2}{2}$$

$$V_0 - V_B = \frac{B\omega\ell^2}{2}$$

$$\therefore V_A = V_B \therefore V_A - V_B = 0$$

Q2

$$\phi = Mi = BA$$

$$\Rightarrow Mi = \frac{\mu_0 i}{2\,b}\pi a^2$$

$$\therefore M = \frac{\mu_0 \pi a^2}{2\,b}$$

Q3

$$I = 3t + 8$$

$$\varepsilon = 12mV$$

$$|\varepsilon| = L\left|\frac{dI}{dt}\right|$$

$$12 = L \times 3$$

$$L = 4mH$$

Q4

$$(Emf)_{\text{induced}} = -L\frac{di}{dt}$$

In magnitude form,

$$|Emf_{ind}| = \left|(-)L\frac{di}{dt}\right|$$

$$\Rightarrow 0.1 = \frac{(L)[+2-(-2)]}{0.2}$$

$$\Rightarrow L = \frac{0.1 \times 0.2}{4} = 5mH$$

Q5

Solutions

$\epsilon = NB\ell v$

$i = \frac{\epsilon}{R} = \frac{NB\ell v}{R}$

$F = N(i\ell B) = \frac{N^2B^2\ell^2 v}{R}$

$\mathrm{W} = \mathrm{F} \times \ell = \frac{\mathrm{N}^2\ \mathrm{B}^2 \ell^3}{\mathrm{R}}\left(\frac{\ell}{\mathrm{t}}\right)$

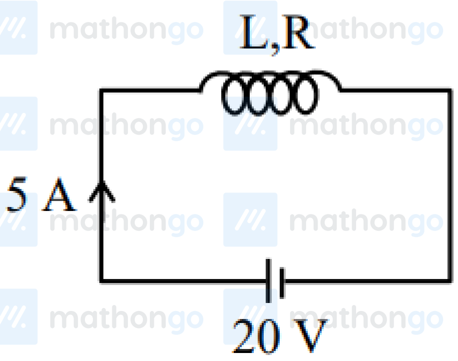
$\mathrm{A} = \ell^2$

$\mathrm{W} = \frac{(10 \times 10)(0.5)^2 \times \left(3.6 \times 10^{-3}\right)^2}{100 \times 1}$

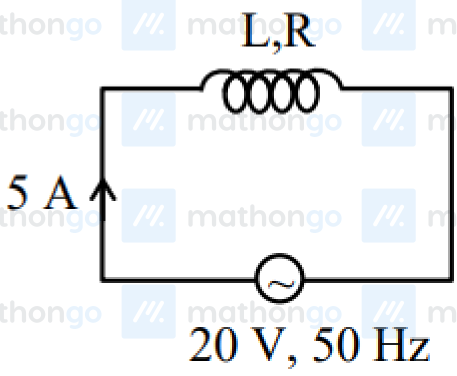
$\mathrm{W} = 3.24 \times 10^{-6}\ \mathrm{J}$

Q6

Case-I:



$i = \frac{20}{R} \Rightarrow R = 4\Omega$



Case-II:

$i = \frac{20}{Z}$

$4 = \frac{20}{\sqrt{R^2 + X_L^2}} \Rightarrow \sqrt{R^2 + X_L^2} = 5$

$R^2 + X_L^2 = 25 \Rightarrow X_L = 3\Omega$

$L = \frac{3}{2\pi f} = \frac{1}{2 \times 50} = \frac{1000}{100} mH$

$L = 10mH$

Q7

At $t = 10sec$ complete loop is in magnetic field therefore no change in flux

$e = \frac{d\phi}{dt} = 0$

$e = 0$ for complete loop